

STA 209: INTRODUCTION TO TIME SERIES ANALYSIS

Extra Credit

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Disclosure

GPT-5.4 (High) on Codex was used to create the **YAML** portion and some **LaTeX** code to format the text and equations. Page-formatting code was also provided by Derin Gezgin.

Problem 1

PROBLEM 1

Consider a time series defined by

$$X_t = \delta + X_{t-1} + W_t,$$

for $t = 1, 2, \dots$ and $X_0 = 0$, where W_t is a white noise with mean 0 and variance 1.

Problem 1 (a)

PROBLEM 1 (A)

Identify the model.

The model

$$X_t = \delta + X_{t-1} + W_t$$

is a random walk with drift δ . That is, each new value equals the previous value, plus a constant drift term δ , plus a white-noise term W_t .

Problem 1 (b)

PROBLEM 1 (B)

Show that X_t can be written as $X_t = t\delta + \sum_{j=1}^t W_j$.

Starting from

$$X_t = \delta + X_{t-1} + W_t,$$

substitute recursively:

$$\begin{aligned} X_t &= \delta + X_{t-1} + W_t \\ &= \delta + (\delta + X_{t-2} + W_{t-1}) + W_t \\ &= 2\delta + X_{t-2} + W_{t-1} + W_t \\ &= 2\delta + (\delta + X_{t-3} + W_{t-2}) + W_{t-1} + W_t \\ &= 3\delta + X_{t-3} + W_{t-2} + W_{t-1} + W_t \\ &\vdots \\ &= t\delta + X_0 + \sum_{j=1}^t W_j. \end{aligned}$$

Since the problem gives $X_0 = 0$, this becomes

$$X_t = t\delta + \sum_{j=1}^t W_j.$$

Problem 1 (c)**PROBLEM 1 (C)**Find the mean and ACVF of X_t .

From part (b),

$$X_t = t\delta + \sum_{j=1}^t W_j.$$

First, compute the mean:

$$\begin{aligned} E(X_t) &= E\left(t\delta + \sum_{j=1}^t W_j\right) \\ &= E(t\delta) + \sum_{j=1}^t E(W_j) \\ &= t\delta + \sum_{j=1}^t 0 \\ &= t\delta. \end{aligned}$$

Hence,

$$\mu_X(t) = t\delta.$$

Now, compute the auto-covariance function. For integers h such that $t+h \geq 0$,

$$\gamma_X(t, h) = \text{Cov}(X_{t+h}, X_t) = E[(X_{t+h} - \mu_{t+h})(X_t - \mu_t)].$$

Since

$$\mu_{t+h} = (t+h)\delta \quad \text{and} \quad \mu_t = t\delta,$$

and from part (b),

$$X_{t+h} = (t+h)\delta + \sum_{j=1}^{t+h} W_j \quad \text{and} \quad X_t = t\delta + \sum_{k=1}^t W_k,$$

it follows that

$$\begin{aligned} X_{t+h} - \mu_{t+h} &= \left((t+h)\delta + \sum_{j=1}^{t+h} W_j \right) - (t+h)\delta \\ &= \sum_{j=1}^{t+h} W_j, \end{aligned}$$

and

$$\begin{aligned} X_t - \mu_t &= \left(t\delta + \sum_{k=1}^t W_k \right) - t\delta \\ &= \sum_{k=1}^t W_k. \end{aligned}$$

Therefore,

$$\begin{aligned}\gamma_X(t, h) &= \mathbb{E} \left[\left(\sum_{j=1}^{t+h} W_j \right) \left(\sum_{k=1}^t W_k \right) \right] \\ &= \sum_{j=1}^{t+h} \sum_{k=1}^t \mathbb{E}(W_j W_k).\end{aligned}$$

Because W_t is white noise with variance 1,

$$\mathbb{E}(W_j W_k) = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

Now, let us evaluate the expression case-by-case. If $h \geq 0$, then

$$\gamma_X(t, h) = \sum_{j=1}^{t+h} \sum_{k=1}^t \mathbb{E}(W_j W_k).$$

Here, the only matching indices are

$$j = k = 1, 2, \dots, t,$$

so there are t nonzero terms, each equal to 1. Therefore,

$$\gamma_X(t, h) = t.$$

If $-t \leq h < 0$, then $t + h < t$, so the only matching indices are

$$j = k = 1, 2, \dots, t + h.$$

Thus, there are $t + h$ nonzero terms, each equal to 1. Therefore,

$$\gamma_X(t, h) = t + h.$$

Hence, the ACVF is

$$\gamma_X(t, h) = \begin{cases} t, & h \geq 0, \\ t + h, & -t \leq h < 0. \end{cases}$$

Problem 1 (d)

PROBLEM 1 (D)

Is this time series stationary? Justify.

No, X_t is not stationary. For (weak) stationarity, two conditions are required:

1. The mean function must not depend on time t .
2. The ACVF must depend only on lag h , not on time t .

But here,

$$\mu_X(t) = t\delta,$$

which depends on t unless $\delta = 0$. So, in general, the mean is not constant over time. Therefore, the process already fails the first condition for weak stationarity. Hence, X_t is not stationary.